

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide $K_4Fe(CN)_6$. After filtering, the precipitate is dried and weighed. The reactions which occur are $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$ $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$ If a 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight $Zn_2Fe(CN)_6 = 342.6$.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

Let's solve the problem step by step :

We are given :

- A 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$.
- We are to find the **percentage of Zn** in the original rock sample .
- We are given :
- Atomic weight of Zn = 65.4
- Molecular weight of $Zn_2Fe(CN)_6 = 342.6$

We are told that ZnS is dissolved by HCl and then Zn^{2+} is precipitated as $Zn_2Fe(CN)_6$.

From the chemical equations :

- $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$
- $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$

So, 2 moles of $ZnCl_2$ produce 1 mole of $Zn_2Fe(CN)_6$.

Thus, 1 mole of $Zn_2Fe(CN)_6$ contains **2 moles of Zn**.

So, we will calculate the mass of Zn in 0.969 g of $Zn_2Fe(CN)_6$, then use that to find the % Zn in the original rock sample .

Step 1 : Moles of $Zn_2Fe(CN)_6$ formed

Molar mass of $Zn_2Fe(CN)_6 = 342.6$ g/mol

Moles of $Zn_2Fe(CN)_6 = \text{mass} / \text{molar mass} = 0.969 / 342.6$

= 0.002829 moles

Step 2 : Moles of Zn in the precipitate

Each mole of $Zn_2Fe(CN)_6$ contains 2 moles of Zn

So, moles of Zn = $2 \times 0.002829 = 0.005658$ mol

Step 3 : Mass of Zn

Mass of Zn = moles $\times$ atomic mass = 0.005658×65.4

= 0.3704 g (approximately)

Step 4 : Percentage of Zn in the rock sample

Total sample mass = 2 g

% Zn = $(\text{mass of Zn} / \text{total sample mass}) \times 100$
= $(0.3704 / 2) \times 100$
= 18.52 %

So, the percentage of Zn in the sample is approximately **18.5 %**

Looking at the choices :

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Closest value to 18.5 % is **19 %** (D) though 18 % is also close .

But let's verify if we did it accurately .

Recalculate step